Kedros Analytics

WORKING CAPITAL IN INVENTORY

How Much Cash Is Stock Holding?

Inventory is one line on the balance sheet. Underneath, some units hold too much and others too little, and the totals hide both.

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THE OPPORTUNITY

Stable totals can hide two expensive mistakes

Most companies never put a number on how much working capital inventory should tie up.

THE BALANCE SHEET

“Our inventory levels feel about right.”

One line each for inventory, receivables and payables. While totals look stable, nobody asks whether each unit runs the right policy.

THE POLICY PER UNIT

Too much, too little, or about right?

Too much pays the cost of capital on frozen stock. Too little loses sales and goodwill. Firm-wide, the two can cancel out on paper.

WHAT IT IS

From “it feels about right” to a policy label per unit

The cash cycle and the carrying-versus-shortage trade-off, applied to each business unit.

1

Your financials

Per unit: credit sales, COGS, inventory, receivables and payables balances, cost of capital, storage rate, stockout risk.

2

Validate and compute

Bad rows are rejected. The rest get cycle days, financing gap, costs and an optimal inventory level I*.

3

A policy label

Each unit is flexible, restrictive or near-optimal, compared with where it should be.

Cash conversion cycle = days in inventory + days in receivables − days in payables.

WHAT YOU GAIN

Six ways it pays off for a small business

1 How long cash is trapped

The cash conversion cycle is a financing requirement with a dollar cost, not “bad operations”.

2 The financing gap in dollars

CCC × (COGS ÷ 365) shows what trade credit leaves to borrowing or a liquidity reserve.

3 An optimal level, not a current one

Carrying costs against shortage costs give an optimal investment I* to compare with actual stock.

4 A policy label per unit

Flexible, restrictive or near-optimal, because averages across units hide exactly this.

5 Bad records stay out

Negative COGS, blank rates and duplicate IDs are rejected with a stated reason.

6 One workbook for the CFO’s office

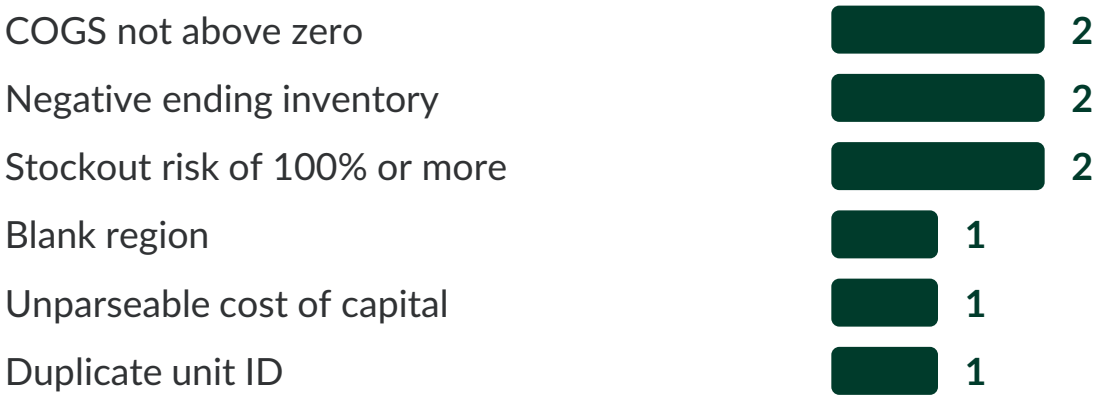
Color-coded Excel with off-policy rows highlighted and a per-segment rollup.

Highlighted: what firm-wide totals can never show.

CATCHING BAD RECORDS

Nine rows rejected before they could skew a cycle

151 rows read, 142 passed. Each rejection has a stated reason.



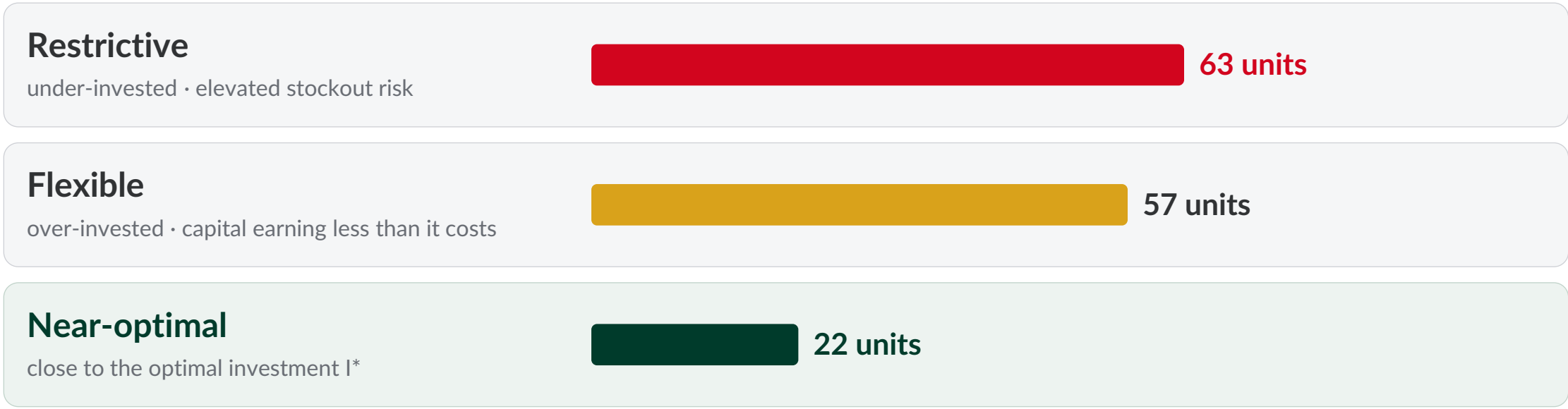
A negative COGS flips the sign of days in inventory and corrupts every average built on it.

Synthetic demo data. Rejected rows by failure reason.

EXAMPLE RESULTS

Unremarkable averages hide a split policy

Averages look fine: 65.6 days in inventory, 48.0 in receivables, 44.0 in payables, a 69.7-day cash cycle.



Only 22 of 142 units are near optimal. Cash cycles run from 185.8 days to −14.8, and seven units run a negative cycle.

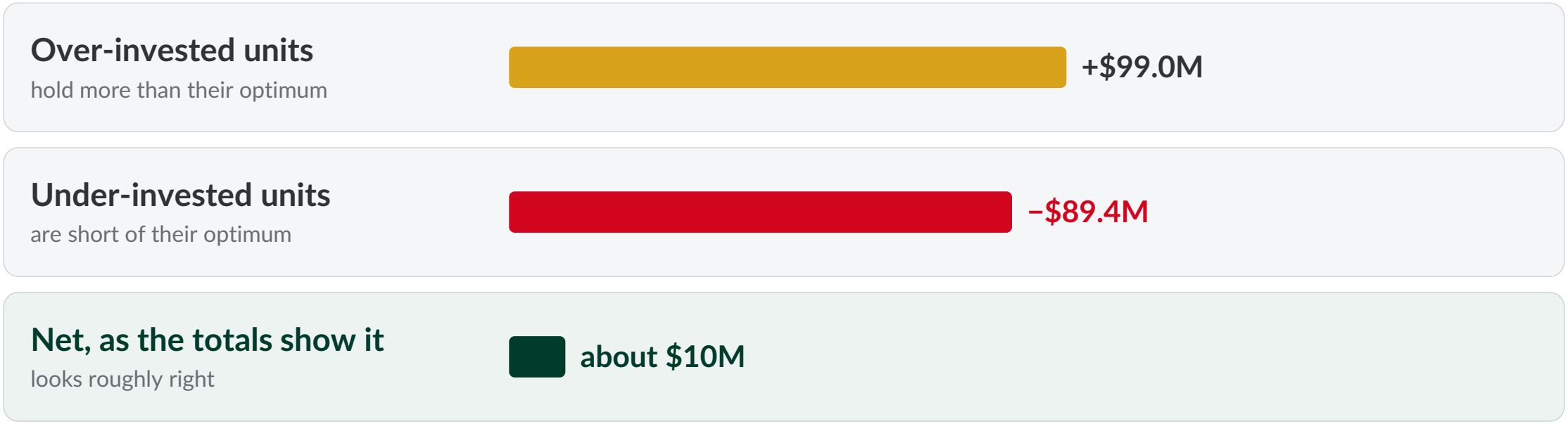
Synthetic demo data. Bars scaled to unit counts.

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THE TRAP IN THE TOTALS

Over- and under-investment almost cancel out

Distance from each unit's optimal inventory, added up across the 142 clean units.



120 of 142 units run the wrong policy while the consolidated balance sheet shows nothing wrong.

Synthetic demo data. Bars scaled to dollars.

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THE SEGMENT ROLLUP

Where to look first

Unit counts, cash cycle, financing gap and policy mix by business segment.

FOOD & BEVERAGE

$85.1M

Largest financing gap.

23 units and an 80.6-day average cash cycle. Ask why so little is covered by trade credit.

CONSUMER ELECTRONICS

0 of 16

Units near optimal.

10 restrictive and 6 flexible: no policy anchored to the carrying-versus-shortage trade-off.

PHARMACEUTICALS

$77.7M

Second-largest gap, short cycle.

A 57.9-day cycle, yet 24 high-volume units. A short cycle is not a small funding need.

BUSINESS IMPACT

A funding need nobody decided to take on

Demo company: 142 clean business units.

$508.7M

of cash cycle that trade credit does not cover: borrowed or held in reserve.

About $99M could be released

Inventory above optimum in the 57 over-invested units.

$89.4M of shortfall

In the 63 under-invested units, where stockout exposure quietly builds.

Synthetic demo data. Educational illustration, not a treasury system.

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GET STARTED

What to do this week

Five steps from last year's figures to a policy per unit.

1

Gather

Per unit: credit sales, COGS, balances, cost of capital, storage rate, stockout risk.

2

Validate

Check the rejection report first. Negative COGS and duplicate IDs are common.

3

Sort

Sort by recommended adjustment. Amber releases capital; red shows stockout exposure.

4

Compare

Judge each unit against its optimum I*, not the firm-wide total.

5

Repeat

Re-run quarterly. Sales mix, supplier terms and interest rates move the optimum.

Do not stop at the totals: in the demo they looked fine while 120 of 142 units were off-policy.

Open-source demo: github.com/MattProtopapas/Inventories-Finance-Demo

KEEP IT HONEST

The limits worth keeping in mind

The flexible and restrictive labels are a screening signal, not a verdict.

1 Shortage cost is the soft spot

Carrying costs are observable. Lost sales and goodwill must be estimated, and I* moves with them.

2 A snapshot built on averages

Seasonality hides in annual balances. A near-optimal unit can swing both ways in a year.

3 A long cycle is not always a problem

Some businesses fund flexible policies knowingly. The framework prices the choice.

4 Illustrative parameters

Cost rates, risk percentages and the near-optimal band are not calibrated to any real firm.

HOW KEDROS ANALYTICS CAN HELP

Try it on your own data, with a free proof of concept

Business questions first, AI as a tool and not a hype machine, and results you can act on straight away.

1

Validate

Check your management-accounting extract for negative costs, impossible balances and duplicate units.

2

Measure the cycle

Compute cash cycle, financing gap and an optimal inventory level for each unit.

3

Rebalance and track

Release capital where stock is excessive, protect service where it is thin, and re-run quarterly.

Free PoC: tailored to your requirements and your company’s analytics maturity.

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Ready to see how much cash is wasted in excess inventory?

Let’s spend thirty minutes on your inventory and cash-cycle figures. No commitment, no jargon, and we might even plan a free PoC.

Book a free call

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